

Week 3 Recitation

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ECON8711 (Micro 1A)

Sep 08, 2023

Please let me know your thoughts and suggestions
till every recitation class:

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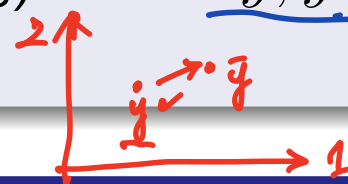
Monotonicity and Nonsatiation

Definition (Monotonicity)

"More is better"

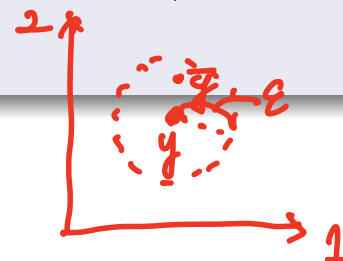
Preference $\succeq$ is monotone (i.e., nondecreasing) if for $\forall y, \bar{y} \in \mathbb{R}_+^N$ such that $\bar{y} \geq y$, then $\bar{y} \succeq y$.

$$\bar{y}_i \geq y_i \quad i=1, 2, \dots, N$$



Definition ((Local) Non-Satiation)

Preference $\succeq$ is Locally Non-Satiated if for $\forall y \in \mathbb{R}_+^N$, $\forall \varepsilon \in \mathbb{R}_{++}$, there exists $\bar{y}$ s.t. $\|y - \bar{y}\| \leq \varepsilon$ with $\bar{y} \succeq y$.



Monotonicity ✓

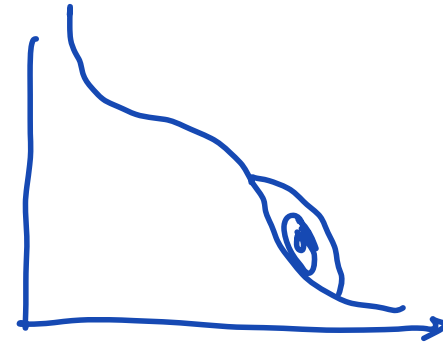
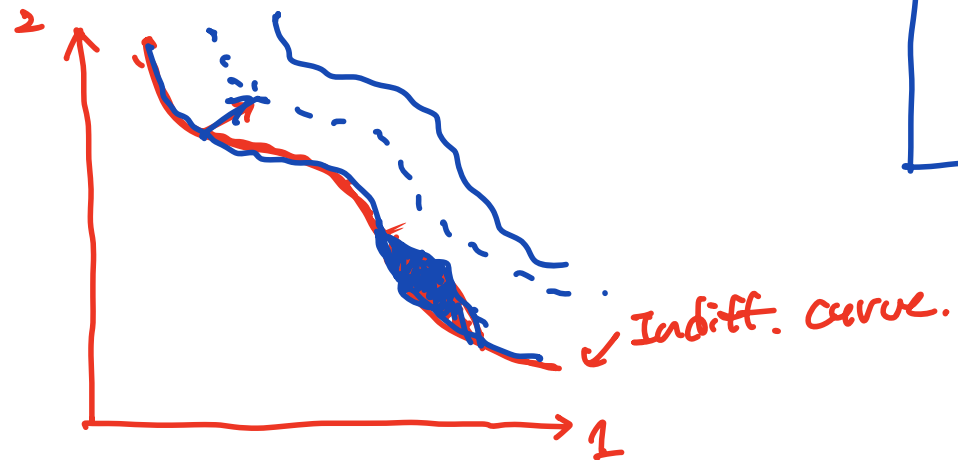


Local Non-satiation

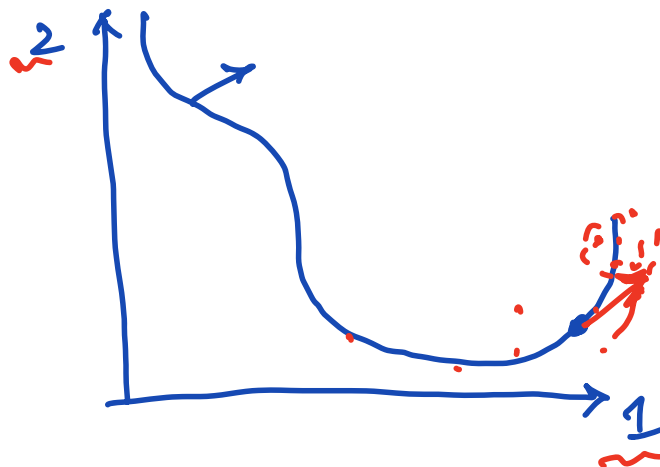
Monotonicity and Nonsatiation

Examples:

① When LNS is violated:



② When LNS holds, but not monotone:



Monotonicity and Nonsatiation

Proposition

Monotonicity of a preference implies the Non-satiability of the preference.

Proof.

Suppose $\succsim$ is monotone. ✓

Take $y \in \mathbb{R}_+^N$ and $0 < \varepsilon < 1$.

Set $\bar{y} \equiv (y_1 + \frac{\varepsilon^2}{\sqrt{N}}, y_2 + \frac{\varepsilon^2}{\sqrt{N}}, \dots, y_N + \frac{\varepsilon^2}{\sqrt{N}})$

By monotonicity, $\bar{y} \succsim y$.

Then $\|\bar{y} - y\| = N \cdot \frac{\varepsilon^2}{\sqrt{N}} = \varepsilon^2 \leq \varepsilon$

$\succsim$ is non-satiable.



Utility representation

$$\underbrace{x}_{\text{wavy}} \succeq y \iff \underbrace{u(x)}_{\text{wavy}} \geq u(y) \quad \checkmark$$

- ① When Countable set of alternatives: (e.g., Suppose X is finite.)

If the preference $\succeq$ on a countable set X is complete and transitive, then there exists a preference function $u : X \rightarrow \mathbb{R}$ s.t.
 $x \succeq y \iff u(x) \geq u(y)$.

- ② When uncountable set of alternatives: (e.g., Suppose $X = \mathbb{R}$.)
➔ We need some assumption on "Density" between alternatives to connect the orderings to a function. (Continuity)

If $X = \mathbb{R}_+^N$ and preferences $\succeq$ are complete, transitive, and continuous on X , then $\succeq$ can be represented by a utility function u (i.e., there exists a preference function $u : X \rightarrow \mathbb{R}$ s.t.
 $x \succeq y \iff u(x) \geq u(y)$.)

Utility representation

Notes on Monotone[✓] / LNS preference[✓] representation
↳ monotone increasing pref. function.

Convexity of preference & its representation

Definition (Convexity of preference)

Let the bundles $y, \bar{y} \in \mathbb{R}_+^N$ where $y \succeq \bar{y}$. The preference $\succeq$ is convex if for $\forall \lambda \in [0, 1]$, $\lambda y + (1 - \lambda)\bar{y} \succeq \bar{y}$.

Definition (Quasi-concavity)

A function $f : \mathbb{R}^N \rightarrow \mathbb{R}$ is quasi-concave when for $\forall \lambda \in [0, 1]$,

$$f(\lambda x + (1 - \lambda)y) \geq \min\{f(x), f(y)\}$$

Convexity of preference & its representation

Proposition

Suppose that u represents the preference $\succeq$. $x \succeq y \Leftrightarrow u(x) \geq u(y)$
 u is quasi-concave iff the preference $\succeq$ is convex.

Proof.

($\Rightarrow$) Suppose u is quasi-concave.

Take any $x, y \in \mathbb{R}_+^N$.

WLOG, assume $u(x) \geq u(y) \Leftrightarrow x \succeq y$

Then by the quasi-concavity of u , $u(\lambda x + (1-\lambda)y) \geq u(y)$ for $\forall \lambda \in [0, 1]$.

Take any $\lambda \in [0, 1]$.

Thus, the pref. $\succeq$ is convex.

($\Leftarrow$) Suppose $\succeq$ is convex.

That is, taking any $x, y \in \mathbb{R}_+^N$, $\lambda \in [0, 1]$, $\lambda x + (1-\lambda)y \succeq y$.

s.t. $x \succeq y$ (WLOG)

$u(\lambda x + (1-\lambda)y) \geq u(y)$ and $u(x) \geq u(y)$.

by definition

"No-Worse-Than Set"
 : upper-contour set

$\uparrow$ set convexity $\Leftrightarrow u$ is quasi-concave.

How to Optimize - Lagrangian

Problem:

$$\begin{aligned} \max_{y \in \mathbb{R}^N} \quad & \sum_{n=1}^N \alpha_n \ln(y_n) \\ \text{s.t.} \quad & p \cdot y \leq m \end{aligned}$$

Handwritten notes: A blue checkmark is above the summation index N . A blue checkmark is below the $\ln(y_n)$ term. A blue checkmark is to the right of the constraint $p \cdot y \leq m$. A blue arrow points from the constraint to the handwritten equation $\sum_{n=1}^N p_n y_n \leq m$.

Lagrangian:

$$\mathcal{L}(y, \mu) = \sum_{n=1}^N \alpha_n \ln(y_n) + \underbrace{\mu \left(m - \sum_{n=1}^N p_n y_n \right)}_{\text{wavy line under } \mu}$$

where $\mu \geq 0$

➔ Take first-order conditions and try to solve the problem

Constraint and Non-satiation

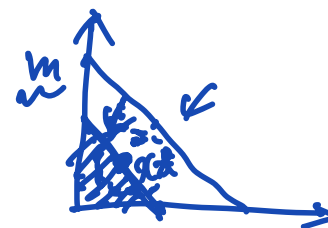
Proposition

If the preference $\succeq$ satisfies Local Non-Satiation and u represents $\succeq$, then the constraint binds at the optimum.

Proof.

$$p \cdot x^* = m, \text{ not } p \cdot x^* < m \checkmark$$

$$p \cdot x \leq m$$



Suppose $\succeq$: LNS ✓

~~max~~ Suppose BWOC, x^* solves

$$\max u(x) \text{ s.t. } p \cdot x \leq m, \text{ but } p \cdot x^* < m \checkmark$$

"Budget($x; p, m$)"

$$\text{Set } \varepsilon > 0 \text{ s.t. } B(x^*, \varepsilon) \subseteq \text{Budget}(p, m).$$

$$\exists y \in B(x^*, \varepsilon) \text{ s.t. } y \succ x^* \checkmark$$

$$u(y) > u(x^*)$$

This contradicts that x^* is optimal

$$\mathcal{L} = \sum_{n=1}^N \alpha_n \ln(y_n) + \mu \left(m - \sum_{n=1}^N p_n y_n \right)$$

$$\frac{\partial \mathcal{L}}{\partial y_n} = \alpha_n \frac{1}{y_n} - \mu p_n = 0 \quad \Leftrightarrow \quad \mu p_n = \frac{\alpha_n}{y_n} \quad \Leftrightarrow \quad \mu p_n y_n = \alpha_n \dots \textcircled{*}$$

$$\frac{\partial \mathcal{L}}{\partial \mu} = m - \sum_{n=1}^N p_n y_n = 0 \quad \Leftrightarrow \quad m = \sum_{n=1}^N p_n y_n$$

$$\mu m = \sum_{n=1}^N \alpha_n$$

$$\mu = \frac{1}{m} \sum_{n=1}^N \alpha_n \quad \checkmark$$

$$\begin{aligned} \textcircled{*}: \quad y_m &= \frac{\alpha_m}{\mu p_m} = \frac{\alpha_m}{\frac{p_m}{m} \sum_{n=1}^N \alpha_n} \\ &= \frac{1}{p_m} \underbrace{\frac{m \alpha_m}{\sum_{n=1}^N \alpha_n}}_{\text{Demand}} \end{aligned}$$